Mittuniversitetet
MID SWEDEN UNIVERSITY

Försättsblad tentamen / Examination cover

2707183

Anonymitetskod / Anonymous code		
C - 0021 - OKX		
Kurskod / Course code: MA074G	Provkod/ Test code: T300	Tentamensdatum / Examination date: 2025-09-26
Kursnamn / Course name: Matematik GR (A), Matematisk statistik och linjär algebra		
Provnamn / Test name: Skriftlig tentamen, Linjär algebra Casio fx-CG50		

Skriv din anonymitetskod på varje inlämnat papper
Write your anonymous code on each sheet submitted

Sätt ett kryss (x) för varje inlämnad uppgift
Use an x to indicate which questions has been submitted

Markera nedan med X / Mark below with an X		Poäng / Credit	Lärarens anteckningar / Teacher's notes	Markera nedan med X / Mark below with an X		Poäng / Credit	Lärarens anteckningar / Teacher's notes
1	X	4		16			
2	X	4.5		17			
3	X	3,5		18			
4	X	4		19			
5	X	0		20			
6				21			
7				22			
8				23			
9				24			
10				25			
11				26			
12				27			
13				28			
14				29			
15				30			
Poängsumma / Points		16	Betyg / Grade C	Lärarsign./ Teachers sign		J Hodén	

Fylls i av tentamensvakt / To be filled in by the invigilator

Antal lösa blad/ No. of sheets submitted	Inlämnad tentamen / Submitted exam	Leg kontroll / Control identification	Sign. tentamensvakt / Sign. invigilator
8	O	✓	

2707183

Försättsbladet skall alltid lämnas in även om ingen uppgift behandlats
Examination cover should always be submitted even if no questions are answered

Utskriven 2025-09-25 kl. 15:01:58

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Anonymitetskod/ Anonymus code: C-0021-0KX

Sidnr / Page no: 1

Lärarens anteckning / Teachers note:

1a) $\begin{bmatrix} 6 & 1 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 9 & 0 & 2 \\ -1 & 2 & 1 & 1 \end{bmatrix} =$

AC: 2×2 2×4 = lika, det fungerar!

Svaret blir en 2×4 matris.

$$\begin{bmatrix} 6-1 & 54+2 & 0+1 & 12+1 \\ 2+1 & 18-2 & 0-1 & 4-1 \end{bmatrix} = \begin{bmatrix} 5 & 56 & 1 & 13 \\ 3 & 16 & -1 & 3 \end{bmatrix}$$

2×4 1R

CB: $\begin{bmatrix} 1 & 9 & 0 & 2 \\ -1 & 2 & 1 & 1 \end{bmatrix} \begin{bmatrix} -1 & 2 \\ -1 & 0 \end{bmatrix}$

2×4 2×2 ej lika = går ej utföra. 1R

det(C): går ej, den är ej kvadratisk 1R

det(A-B): $\begin{bmatrix} 6 & 1 \\ 2 & -1 \end{bmatrix} - \begin{bmatrix} -1 & 2 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} 7 & -1 \\ 3 & -1 \end{bmatrix}$

$\det(A-B) = 7\cdot -1 - 3\cdot -1 = -7+3 = -4$ 1R

$\det(3A^4)$: pga 2×2 matris! $\begin{bmatrix} 6 & 1 \\ 2 & -1 \end{bmatrix}_{2\times 2} = 6\cdot -1 - 2\cdot 1 = -6-2 = -8$

$\det 3A^4 = 3^2 \cdot (-8)^4 = 9 \cdot 4096 = 36864$ 1R

4p

Skriv ej på baksidan! Do not write on the other side!

Lärarens anteckning / Teachers note:

1

2a) $\vec{v}_1 \perp \vec{v}_2 \iff \vec{v}_1 \cdot \vec{v}_2 = 0 \Rightarrow$

$\begin{pmatrix} a \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 2 \end{pmatrix} = 2a+2 = 0 \iff a = -1$

Svar: $a = -1$ så är $\vec{v}_1 \perp \vec{v}_2$

1.5

2b) $\dfrac{v_1 \cdot v_2}{\|v_1\| \|v_2\|} = \cos\varphi \Rightarrow \dfrac{\begin{pmatrix} -\sqrt{3} \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 2 \end{pmatrix}}{\sqrt{(-\sqrt{3})^2+1^2}\ \sqrt{2^2+2^2}} =$

$\dfrac{-\sqrt{3}\cdot 2 + 2\cdot 1}{\sqrt{4}\ \sqrt{8}} = \dfrac{2-2\sqrt{3}}{4\sqrt{2}}$

$\cos^{-1}\left(\dfrac{2-2\sqrt{3}}{4\sqrt{2}}\right) = 105^\circ$ Svar: 105°

1

2c) $\vec{u}_1 \times \vec{u}_2$ = en vektor som är $\perp \vec{u}_1 \& \vec{u}_2$.

$\begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} \times \begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix} = \begin{pmatrix} -4-6 \\ -(8-3) \\ 4-(-1) \end{pmatrix} = \begin{pmatrix} -10 \\ -5 \\ 5 \end{pmatrix}$

$\vec{e} = \dfrac{1}{\|\vec{u}_1 \times \vec{u}_2\|}\begin{pmatrix} -10 \\ -5 \\ 5 \end{pmatrix} = \dfrac{1}{\sqrt{-10^2+-5^2+5^2}}\begin{pmatrix} -10 \\ -5 \\ 5 \end{pmatrix} = \dfrac{1}{5\sqrt{6}}\begin{pmatrix} -10 \\ -5 \\ 5 \end{pmatrix} =$

$\begin{pmatrix} -\frac{\sqrt{6}}{3} \\ -\frac{\sqrt{6}}{6} \\ \frac{\sqrt{6}}{6} \end{pmatrix}$ Svar: $\begin{pmatrix} -\frac{\sqrt{6}}{3} \\ -\frac{\sqrt{6}}{6} \\ \frac{\sqrt{6}}{6} \end{pmatrix}$

This is the unit vector perpendicular to the span of $\vec{u}_1$ and $\vec{u}_2$. But what we need is **all** such vectors. They can be easily obtained if we multiply it by $k \in \mathbb{R}$.

Lärarens anteckning / Teachers note:

2d) $\|\vec{u}_1 \times \vec{u}_2\| = A_P$

$\vec{u}_1 \times \vec{u}_2 = \begin{pmatrix}2\\-1\\3\end{pmatrix} \times \begin{pmatrix}1\\2\\4\end{pmatrix} = \begin{pmatrix}-10\\-5\\5\end{pmatrix}$ (fr. 2c)

$\sqrt{(-10)^2+(-5)^2+5^2} = 5\sqrt{6}$ (fr. 2c)

Svar: $5\sqrt{6}$ a.e

2e) $P = (1,1,-5)$ $ax+by+cz=d$

$\vec{u}_1 = \begin{pmatrix}2\\-1\\3\end{pmatrix}$ $\vec{u}_1 \cdot \vec{n} = 0 = \perp$

We don't need $\vec{n}$ here.

$\begin{pmatrix}2\\-1\\3\end{pmatrix} \cdot \begin{pmatrix}a\\b\\c\end{pmatrix} = 2a+(-b)+3c = 2a-b+3c$

$2a-b+3c=0 \Rightarrow$ $a=-1$ $b=1$ $c=1$

You determine the direction of one of vectors located on this plane. This is meaningless here because a plane has infinite such direction. BUT! It has only one perpendicular direction. See below.

$\begin{pmatrix}2\\-1\\3\end{pmatrix} \cdot \begin{pmatrix}-1\\1\\1\end{pmatrix} = -2+(-1)+3 = 0$

$ax+by+cz=d$ $\begin{pmatrix}-1\\1\\1\end{pmatrix}$ insatt:

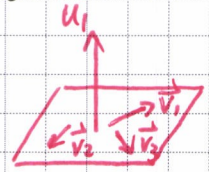
$-x+y+z=d$ sätter in $P(1,1,-5)$

$-1+1+(-5)=d \Rightarrow -5=d$ $d=-5$

$-x+y+z+5=0$

$\vec{r}$ för planet $\perp \begin{pmatrix}-1\\1\\1\end{pmatrix} \perp \vec{u}_1$

Svar: $-x+y+z+5=0$

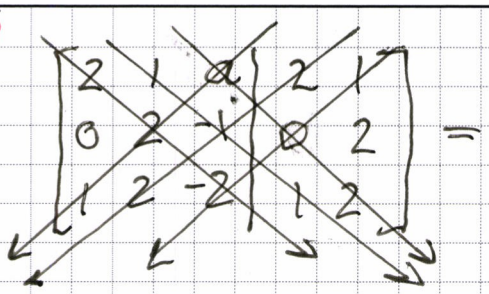


For each plane $\vec{u}_1$ is unique up to a multiplier. But we have infinite $\vec{v}_1, \vec{v}_2, \vec{v}_3 \ldots$ laying on the plane.

4. ✓

Lärarens anteckning / Teachers note:

3a) (1)



$(2\cdot2\cdot-2)+(1\cdot-1\cdot1)+(a\cdot0\cdot2)-(1\cdot0\cdot-2)-(2-1\cdot2)-(a\cdot2\cdot1)$

$-8 + (-1) + 0 - 0 - (-4) - 2a$

$-5-2a=0 \rightarrow -2a=5 \rightarrow 2a=-5$

$a=-\frac{5}{2}$ Om $a \neq -\frac{5}{2}$ så finns invers

$\det(A)=0 \rightarrow$ Invers saknas

$\det(A)\neq 0 \rightarrow$ Invers existerar

Svar: Matris ej inverterbar om $a=-\frac{5}{2}$

Lärarens anteckning / Teachers note:

3b)

2

$$\left[\begin{array}{ccc|ccc} 2 & 1 & 2 & 1 & 0 & 0 \\ 0 & 2 & -1 & 0 & 1 & 0 \\ 1 & 2 & -2 & 0 & 0 & 1 \end{array}\right] \sim \begin{array}{l} r_1 \\ r_2 \\ 2r_3 \end{array}\left[\begin{array}{ccc|ccc} 2 & 1 & 2 & 1 & 0 & 0 \\ 0 & 2 & -1 & 0 & 1 & 0 \\ 2 & 4 & -4 & 0 & 0 & 2 \end{array}\right] \sim \text{ok}$$

$$\begin{array}{l} r_1 \\ r_2 \\ r_3 - r_1 \end{array}\left[\begin{array}{ccc|ccc} 2 & 1 & 2 & 1 & 0 & 0 \\ 0 & 2 & -1 & 0 & 1 & 0 \\ 0 & 3 & -6 & -1 & 0 & 2 \end{array}\right] \text{ok}$$

$$A^{-1} = \begin{bmatrix} \frac{2}{9} & -\frac{2}{3} \\ \frac{1}{9} & \frac{2}{3} \\ \frac{2}{9} & \frac{3}{9} \end{bmatrix}$$

$$\begin{array}{l} r_1 \\ r_2 \\ r_3 - r_2 \end{array}\left[\begin{array}{ccc|ccc} 2 & 1 & 2 & 1 & 0 & 0 \\ 0 & 2 & -1 & 0 & 1 & 0 \\ 0 & 1 & -5 & -1 & -1 & 2 \end{array}\right] \sim \text{ok}$$

$$\begin{array}{l} r_1 \\ r_2 - r_3 \\ r_3 \end{array}\left[\begin{array}{ccc|ccc} 2 & 1 & 2 & 1 & 0 & 0 \\ 0 & 1 & 4 & 1 & 2 & -2 \\ 0 & 1 & -5 & -1 & -1 & 2 \end{array}\right] \sim \text{ok}$$

$$\begin{array}{l} r_1 \\ r_3 \\ r_2 \\ r_3 - r_2 \end{array}\left[\begin{array}{ccc|ccc} 2 & 1 & 2 & 1 & 0 & 0 \\ 0 & 1 & 4 & 1 & 2 & -2 \\ 0 & 0 & -9 & -2 & -3 & 4 \end{array}\right] \sim \text{ok}$$

$$\begin{array}{l} r_1 - r_2 \\ r_2 \\ \cdot\, r_3 \end{array}\left[\begin{array}{ccc|ccc} 2 & 0 & -2 & 0 & -2 & 2 \\ 0 & 1 & 4 & 1 & 2 & -2 \\ 0 & 0 & -9 & -2 & -3 & 4 \end{array}\right] \sim \text{ok}$$

$$\begin{array}{l} 2r_1 \\ r_2 \\ r_3 \end{array}\left[\begin{array}{ccc|ccc} 4 & 0 & -4 & 0 & -4 & 4 \\ 0 & 1 & 4 & 1 & 2 & -2 \\ 0 & 0 & -9 & -2 & -3 & 4 \end{array}\right] \sim$$

$$\begin{array}{l} r_1 + r_2 \\ r_2 \\ r_3 \end{array}\left[\begin{array}{ccc|ccc} 4 & 1 & 0 & 1 & -2 & 2 \\ 0 & 1 & 4 & 1 & 2 & -2 \\ 0 & 0 & -9 & -2 & -3 & 4 \end{array}\right]$$

$$\begin{array}{l} r_1 \\ r_2 \\ \frac{1}{9} r_3 \end{array}\left[\begin{array}{ccc|ccc} 4 & 1 & 0 & 1 & -2 & 2 \\ 0 & 1 & 4 & 1 & 2 & -2 \\ 0 & 0 & -1 & -\frac{2}{9} & -\frac{3}{9} & \frac{4}{9} \end{array}\right] \sim$$

$$\begin{array}{l} r_1 \\ r_2 + 4r_3 \\ -1\, r_3 \end{array}\left[\begin{array}{ccc|ccc} 4 & 1 & 0 & 1 & -2 & 2 \\ 0 & 1 & 0 & \frac{1}{9} & \frac{2}{3} & -\frac{34}{9} \\ 0 & 0 & 1 & \frac{2}{9} & \frac{3}{9} & -\frac{4}{9} \end{array}\right] \quad -2 + \frac{16}{9} = -\frac{2}{9}$$

$$r_1 + r_2 \left[\begin{array}{ccc|ccc} 4 & 0 & 0 & \frac{8}{9} & -\frac{8}{3} & \frac{52}{9} \\ 0 & 1 & 0 & \frac{1}{9} & \frac{2}{3} & -\frac{34}{9} \\ 0 & 0 & 1 & \frac{2}{9} & \frac{3}{9} & -\frac{4}{9} \end{array}\right] \quad \frac{20}{9} \quad -\frac{2}{9}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{2}{9} & -\frac{2}{3} & \frac{13}{9} \\ 0 & 1 & 0 & \frac{1}{9} & \frac{2}{3} & -\frac{34}{9} \\ 0 & 0 & 1 & \frac{2}{9} & \frac{3}{9} & -\frac{4}{9} \end{array}\right] \quad \frac{5}{9} \quad -\frac{2}{9}$$

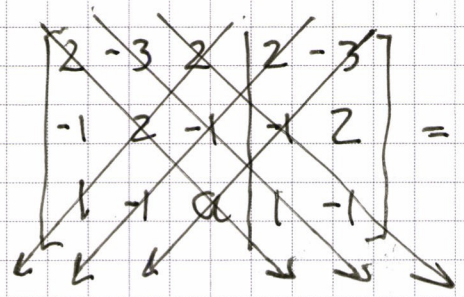
$$\left[\begin{array}{ccc|ccc} 2 & -\frac{2}{3} & \frac{13}{9} & \frac{5}{9} \\ \frac{1}{9} & \frac{2}{3} & -\frac{34}{9} & -\frac{2}{9} \\ \frac{2}{9} & \frac{3}{9} & -\frac{4}{9} & \end{array}\right]$$

3,5

Lärarens anteckning / Teachers note:

4)

$$\left[\begin{array}{ccc|c} 2 & -3 & 2 & 9 \\ -1 & 2 & -1 & -5 \\ 1 & -1 & a & b \end{array}\right]$$



$(2 \cdot 2 \cdot a) + (-3 \cdot -1 \cdot 1) + (2 \cdot -1 \cdot -1) - (-3 \cdot -1 \cdot a) - (2 \cdot -1 \cdot -1) -$

$(2 \cdot 2 \cdot 1) = 4a + 3 + 2 - 3a - 2 - 4 =$

$1a + 5 - 2 - 4 \Rightarrow \quad a - 1 = 0 \quad a = 1$

det (A) = 0 om $a = 1$ → oändligt många

det (A) ≠ 0 om $a \neq 1$ → en exakt lösning

Löser systemet med $a = 1$

nästa blad..

Skriv ej på baksidan! Do not write on the other side!

Lärarens anteckning / Teachers note:

4 forts.)

$$\left[\begin{array}{ccc|c} 2 & -3 & 2 & 9 \\ -1 & 2 & -1 & -5 \\ 1 & -1 & 1 & b \end{array}\right] \sim \begin{matrix} r_1 \\ r_2 \\ r_3 - r_2 \end{matrix} \left[\begin{array}{ccc|c} 2 & -3 & 2 & 9 \\ -1 & 2 & -1 & -5 \\ 2 & -3 & 2 & b-5 \end{array}\right] \sim$$

$b+5$

$$\begin{matrix} r_1 \\ r_2 \\ r_3 - r_1 \end{matrix} \left[\begin{array}{ccc|c} 2 & -3 & 2 & 9 \\ -1 & 2 & -1 & -5 \\ 0 & 0 & 0 & b-5-9 \end{array}\right]$$

$b+5-9 = b-4$

↑ nollrad!

$$b-5-9 = b-14 = 0$$

Om $b-14=0$ så finns oändligt många lösningar

dvs om $b=14$

Om $b-14 \neq 0$ så saknas lösning dvs om

$b \neq 14$ (pga $0+0+0 =$ ett tal)

b can be any reeal number.

4a svar: Om $a \neq 1$ & $b \neq 14$ 1

4b svar: Om $a = 1$ & $b \neq 14$ 2

4c svar: om $a = 1$ & $b = 14$ 1

4

Lärarens anteckning / Teachers note:

5) $L = (4,2,0) + t\begin{pmatrix}2\\-1\\1\end{pmatrix}$ $\quad \vec{r} = \begin{pmatrix}2\\-1\\1\end{pmatrix}$

$$\text{Proj}_{\vec{r}}\vec{QP} = \frac{\vec{QP}\cdot\vec{r}}{\|\vec{r}\|^2}\vec{r}$$

$\vec{QP}$ måste bestämmas, hitta en punkt P på linjen:

$2x - y + z = d$ sätt in känd punkt för att få fram d

$2(4) - 2 + 0 = d \Rightarrow 6 = d$

$2x - y + z - 6 = 0$ hitta en punkt P:

$2(2) - 0 + 2 = 6$ $\quad (2,0,2)$

$$\vec{QP} = (8,0,8)(2,0,2) = \begin{pmatrix}2-8\\0-0\\2-8\end{pmatrix} = \begin{pmatrix}-6\\0\\-6\end{pmatrix}$$

$$\frac{\begin{pmatrix}-6\\0\\-6\end{pmatrix}\cdot\begin{pmatrix}2\\-1\\1\end{pmatrix}}{\sqrt{2^2+(-1)^2+1^2}^2}\begin{pmatrix}2\\-1\\1\end{pmatrix} = \frac{-12+0-6}{6}\begin{pmatrix}2\\-1\\1\end{pmatrix} =$$

Op

$$\frac{-18}{6}\begin{pmatrix}2\\-1\\1\end{pmatrix} = -3\begin{pmatrix}2\\-1\\1\end{pmatrix} = \begin{pmatrix}-6\\3\\-3\end{pmatrix}$$

$\sqrt{(-6)^2+3^2+(-3)^2} = \sqrt{54}$ l.e

Se lösn kursshemsidan

Svar: $3\sqrt{6}$ l.e